VA DIVISION A • GRADES 3–4 • 2025–2026

Astronaut Lander

Competition Project Slides

- Official Rules
- Materials Kit
- Scoring Formula
- Maximize Flight Time
- Human Body in Space
- Competition Tips

Build a device that keeps a marshmallow astronaut safely in its paper cup capsule after a drop from a two-story stairwell.

📋 Event at a Glance

2	45	25	11	100	1/100
Students per team	Min total session	Min build + test	Materials items	Max score pts	Timing precision (s)

GOAL: Build a landing device that keeps the marshmallow astronaut INSIDE the paper cup capsule after dropping from a two-story stairwell. Maximize flight time. Score = flight time + written assessment.

Must use the cup:	The 237ml paper cup is mandatory. Cannot be cut. This is the astronaut capsule.
Marshmallow:	Cannot be taped or wedged into the cup. Exits cup = Tier 2 (−20%).
Written test:	Topics: forces on landing astronauts, effects of space on the human body.
Countdown:	Hold parallel to railing, say "3, 2, 1, drop". Cannot move upward while releasing.

Cup restriction:	Nothing may be built within 10cm directly above the cup lip.
Construction:	25-minute build phase simultaneous with written test.
Announce estimate:	State your estimated flight time to the supervisor during build phase.
Safety:	ANSI Z87+ impact glasses required. Bring your own.

Official Materials Kit

Materials

237 ml paper cup (THE CAPSULE)
Mandatory — must be used, cannot be cut. The marshmallow goes inside.
MUST use

Mini marshmallows (~0.6cm × 0.9cm)
Structural cushioning material.
10

Coffee filter #6 (larger)
Larger parachute material — more drag = slower descent = longer time.
1

Paper straws
Framework to hold cup stable. May be cut.
4 straws

Cardboard piece (~10×13 cm)
Structural base or reinforcement.
1 piece

Masking tape (2.54 cm)
All structural connections.
1 meter

Regular marshmallow (~2.54cm dia × 3.8cm)
The astronaut. Must stay in the cup. Cannot be taped or wedged in.
1

Coffee filter #2 (smaller)
Parachute material. Cone shape traps air efficiently.
1

Rubber bands (#64, #33, #19 — may mix)
Suspension, attachment, or structural. Choose 3 from the assortment.
3 bands

Index cards (8×13 cm)
Structural fins, additional panels. May be cut.
3 cards

String
Parachute shroud lines, structural ties.
1 meter

BRING: ANSI Z87+ impact safety glasses • Writing utensils • Scissors provided at event

Official Scoring Formula

Scoring

Final Score (FS) = Assessment Score (TAS) + Time Score (TTS) • Maximum = 100 points

Term	Formula / Rule	Notes
Assessment Score (TAS)	(Your score ÷ Best assessment score among all teams) × 50	Max 50 pts
Time Score (TTS)	(Your flight time ÷ Longest flight time among all teams) × 50	Max 50 pts
Flight Time	From release until lander gets stuck or comes to complete stop. Bouncing continues the clock.	Recorded to 1/100 sec
Tier 1 (full score)	Lander meets all requirements + marshmallow stays in cup + Spirit of Problem followed.	No penalty
Tier 2 (−20%)	Construction fault OR marshmallow exits the cup on landing.	×0.80 final score
Tier 3 (−30%)	Team did not follow Spirit of the Problem guidelines.	×0.70 final score
Tiebreaker 1	Smallest difference between estimated and actual flight time.	Estimate accurately!
Tiebreaker 2	Correct answers on selected written assessment questions.	Know your content

🔧 Design Strategy — Maximize Flight Time

Strategy

PARACHUTE (coffee filter)

- ◆ #6 filter = larger area = more drag = slower fall
- ◆ Cone shape traps air — don't flatten it out
- ◆ Attach 4+ equally-spaced string lines to edge
- ◆ Lines must be equal length — tilted chute = less drag
- ◆ Cup must hang UPRIGHT throughout descent
- ◆ Test: hold by parachute — does cup hang level?

10 cm RULE + CUP DESIGN

- ◆ NOTHING built within 10cm above the cup lip
- ◆ Attach parachute strings to cup SIDES, not top
- ◆ Marshmallow must sit freely — no tape, no wedging
- ◆ Test: shake gently — does marshmallow stay in?
- ◆ Use rubber bands to suspend cup below a frame
- ◆ Cup opening must face UP throughout the entire fall

WRITTEN TEST TOPICS

- ◆ Gravity: $F = mg$ (9.8 m/s^2 on Earth)
- ◆ Air resistance (drag): opposes motion, grows with speed
- ◆ Impact force: reduced by longer deceleration time
- ◆ g-force: up to 3g on re-entry for astronauts
- ◆ Bone density loss: ~1–2% per month in space
- ◆ Muscle atrophy: affects legs most in microgravity
- ◆ Fluid shift: more blood to head in zero gravity
- ◆ Recovery: weeks to months after long mission

Competition Tips — How to Win

Tips

01

Use the #6 coffee filter first

The #6 is larger than the #2 — more surface area = more drag = slower descent = more flight time. This is your primary parachute.

02

The 10cm rule is critical

Nothing built within 10cm directly above the cup lip. Parachute strings must attach to the cup SIDES or to a frame structure — not crossing over the opening.

03

Keep the cup upright

The cup opening must face up throughout descent or the marshmallow will fall out before landing. Test this by holding your lander by the parachute and checking the cup angle.

04

Go light to stay up longer

Every extra gram increases fall speed. Use minimum materials. The coffee filters are already very lightweight — don't add unnecessary weight with the cardboard or straws.

05

Time your test drop accurately

Tiebreaker 1 is the closest estimate to actual flight time. Use a phone stopwatch during practice. Give a specific number (e.g., "2.4 seconds") not a round guess.

06

Study both test topics

Assessment = 50 pts. Gravity, air resistance, impact force, g-force, and what happens to bones, muscles, and heart in space. Review before competition day.